

HIMANSHU MISHRA

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SUMMARY

Graduate researcher at UBC focusing on **Adaptation AI**, **Model compression**, and **LLM unlearning**. Proficient in **transformer-based architectures**, prompt optimization, and production-grade deployment using Python and cloud platforms, with research interests in continual adaptation, efficient model scaling, and removal of undesired knowledge from LLMs.

EDUCATION

University of British Columbia, Canada | PhD-Track in Computer Science

Sept 2025 - Present

- **Relevant Coursework** - Adaptation & Adaptive AI, Causal Machine Learning, NLP-Interpretation, Commonsense in AI

Indian Institute of Technology Kanpur, India | B. Tech | Minor in Computer Science | CGPA: 8.4/10

Aug 2018 - May 2022

- **Relevant Coursework** - Introduction to Computing, Data Structures & Algorithms, Database Management System, Intro. to Machine Learning, Computer Organization, Applied Probability & Statistics

RESEARCH PROJECTS

RDumb++: Adaptive Reset for Continual Test-Time Adaptation | PyTorch, CIFAR-10-C/100-C

[Paper](#)

- Solved model drift issues in continuous data streams by building a drift-aware adaptive reset framework using entropy and KL-divergence, improving accuracy by **3.8%** and cutting resets by **35%**.
- Achieved **20% faster recovery** and **15% higher stable adaptation** of online ML systems under production domain shifts

Quantization-Aware LLM Unlearning (QUAIL) | Transformers, PyTorch

[Paper](#)

- Developed **QUAIL**, a method that ensures LLMs truly forget targeted data even after **4-bit quantization**. Introduced a logit-space hinge loss and analyzed quantization bucket overlap, evaluated on **MUSE** and **Twitter misinformation** datasets.
- Preserved forgetting effectiveness after low-bit deployment while maintaining reasonable model utility.
- Mitigated catastrophic knowledge recovery caused by quantization in existing unlearning methods.

WORK EXPERIENCE

Senior AI Engineer | Accenture Japan

Nov 2023 - Aug 2025

- Developed and deployed recommendation algorithms that improved user engagement by **18%** across platform features.
- Built computer vision models for content moderation that reduced harmful content exposure across the platform by **32%**.
- Implemented **A/B testing** framework for ML model evaluation, enabling rapid iteration and **40%** faster deployment cycles.
- Optimized database queries and implemented **Redis-based** caching strategies that reduced load times by **60%**.
- Streamlined model inference pipelines, reducing latency by **45%** while preserving model accuracy in production.

Data & AI Engineer | Accenture Japan

Nov 2022 - Oct 2023

- Designed and implemented machine learning models for dynamic pricing that increased host revenue by **15%**.
- Built NLP models using transformer embeddings to extract customer sentiments and satisfaction drivers with **87% accuracy**.
- Developed automated pipelines using **Airflow** to orchestrate feature engineering and model training for scalable deployment.
- Translated business needs into ML solutions and influenced product direction through executive-facing insights.

Machine Learning Researcher | York University Canada

May 2022 - Nov 2022

- Implemented Bayesian optimization with kernel density estimation and L-BFGS tuning, reducing search space to **9.7%**.
- Applied feature engineering, **Random Forest** feature ranking, and **SVM** classification to identify top material candidates.
- Optimized exploration-exploitation and model generalization for materials discovery with enhanced solar cell performance.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, Java, JavaScript/TypeScript.
- **Machine Learning:** NLP, CNN, RNN, Feature Engineering, A/B Testing, Time Series Analysis, Model Deployment.
- **Frameworks & Libraries:** TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn.
- **MLOps:** Docker, AWS SageMaker, MLflow, Apache Airflow, Model Monitoring, CI/CD for ML Pipelines, Git.

AWARDS & ACHIEVEMENTS

- Awarded prestigious **Mitacs Globalink Research Scholarship** for summer internship at York University, Toronto, Canada.
- Ranked among **top 1%** out of 1.5 million candidates in **IIT-JEE Advanced** Competitive Examination 2018.